Assignment-2

ACOUSTIC ZONING ANALYSIS

Graph Analysis of Architectural Structure

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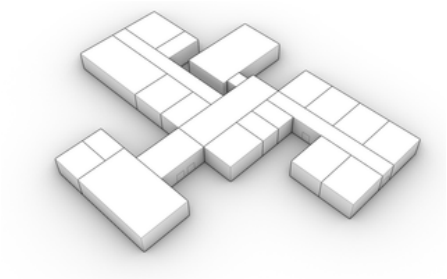
Concept

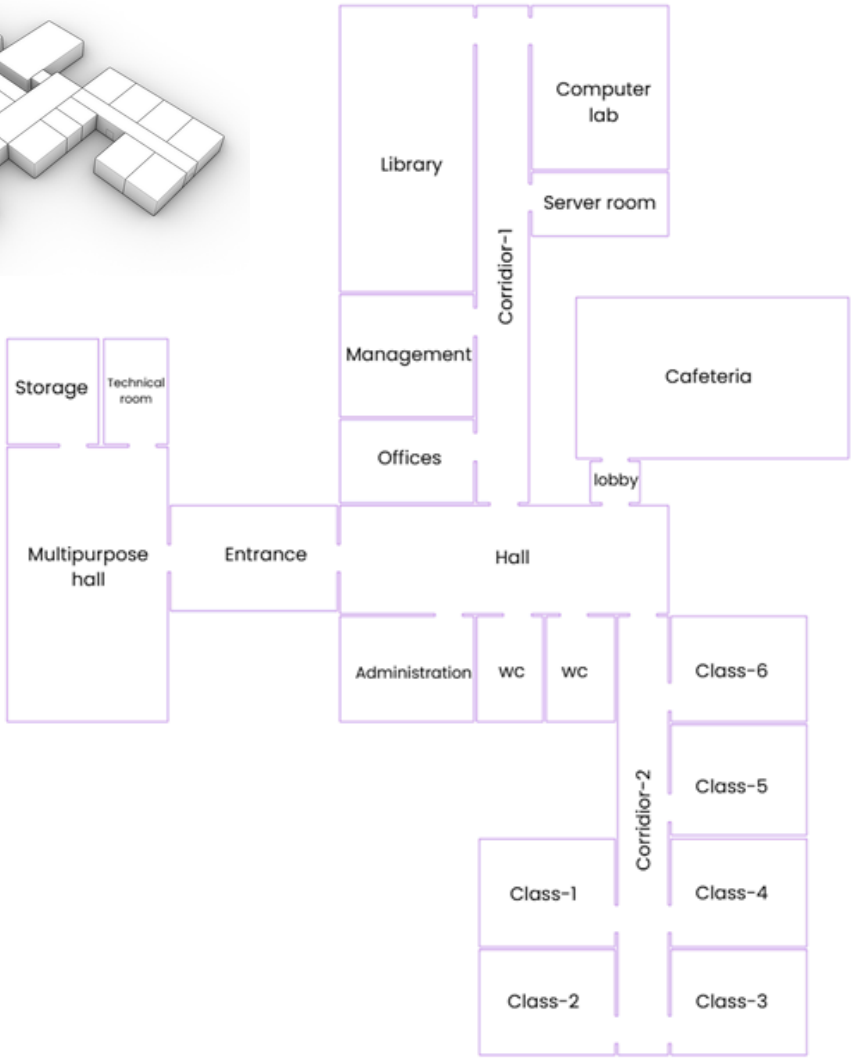
This assignment applies graph-based spatial intelligence to investigate acoustic exposure across a single-floor school building. The central research question is:

"Which spaces in this school are acoustically vulnerable, and which are protected?"

The more walls and rooms it must cross, the more it fades. By modelling the floor plan as a graph, the shortest path between any cell and the nearest noise source (measured in hops) becomes a quantitative proxy for acoustic exposure.

Three noise sources were defined: the Multipurpose Hall, the Entrance, and the Cafeteria. Spaces close to these sources are acoustically exposed (red/yellow). Spaces many hops away like the Library and classrooms are protected (blue).



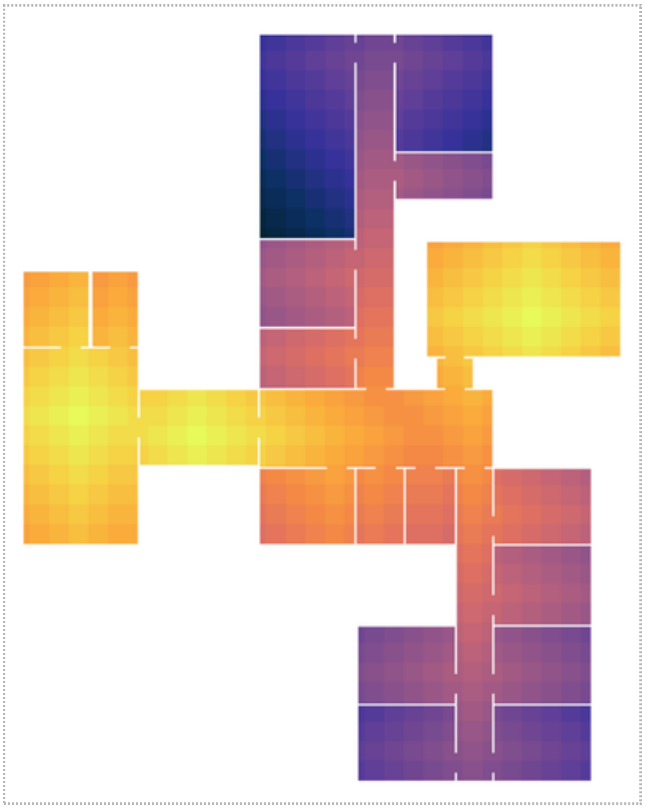


Workflow

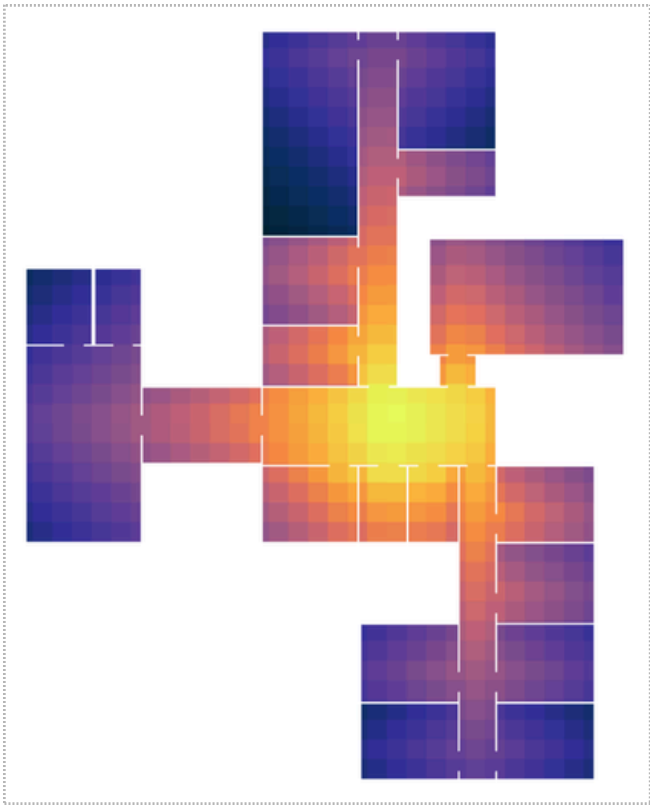
The pipeline's Spatial Intelligence framework: OBJ floor plan, Shell, 2m grid overlay, Topology.Slice(), Graph.ByTopology(), ShortestPath analysis, thermal heatmap. Four metrics were computed:

- **Acoustic Distance :** *Shortest path hops from the 3 noise sources (custom metric)*
- **Closeness Centrality:** *Which cells can reach all others most efficiently?*
- **Betweenness Centrality :** *Which cells are acoustic bottlenecks?*
- **Community Detection :** *Natural acoustic zone clusters*
- **Isovist Analysis:** *Visibility as a proxy for acoustic openness*

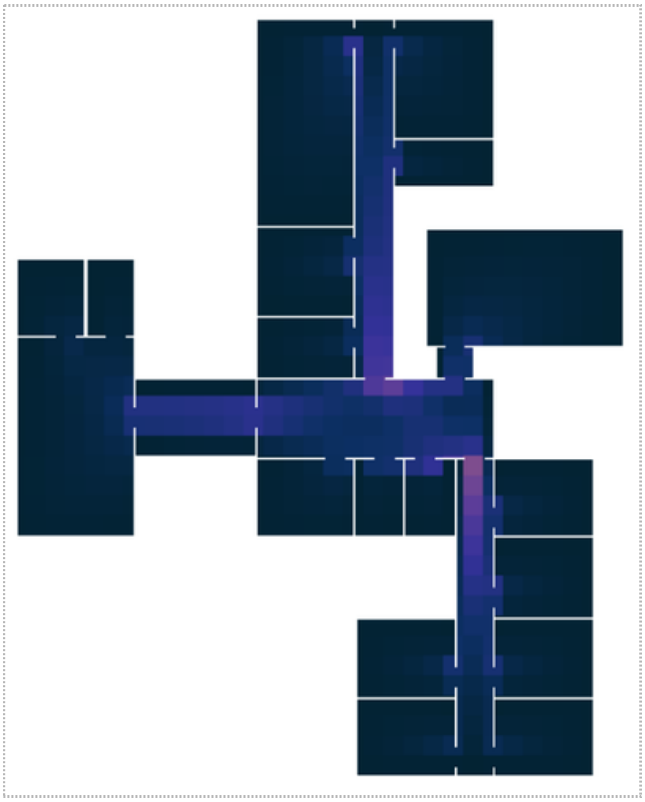
Results



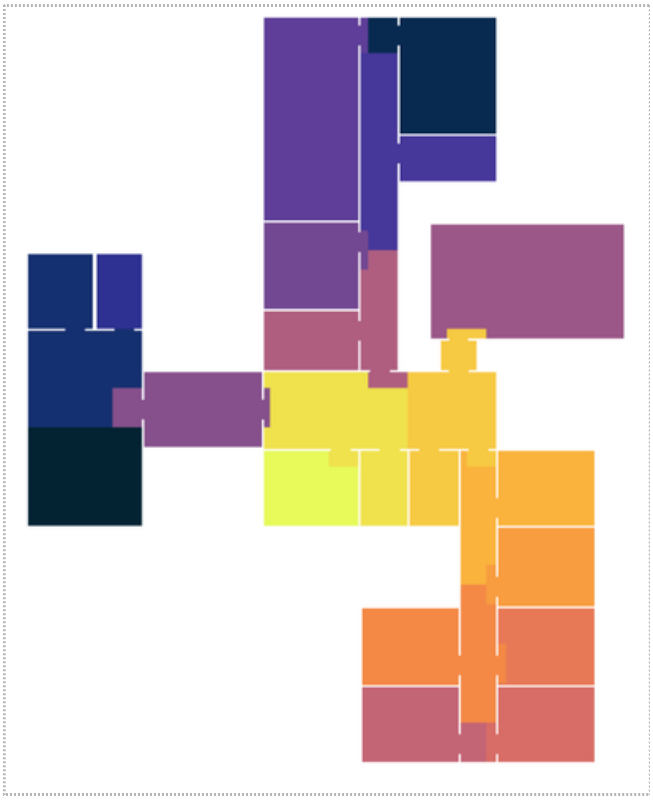
Acoustic Distance
Yellow = noise source | Blue = protected



Closeness Centrality
Yellow = most spatially integrated



Betweenness Centrality
Red = acoustic corridors



Community Detection
Each colour = acoustic zone cluster

Graph Metrics Summary

The following tables present the computed acoustic distance values across all spaces in the school floor plan. Acoustic distance is measured in graph hops — the number of spatial steps separating each cell from the nearest noise source (Multipurpose Hall, Entrance, or Cafeteria). A lower hop count indicates higher acoustic exposure, while a higher hop count indicates greater acoustic protection.

Room name	Acoustic Distance (Hops)
Lobby	5.5
Storage	6.9
Technical Room	8.2
Hall	11.7
Administration	12.8
Corridor-2	13.7
WC	14.5
Corridor-1	18.7
Management	21.5
Class-3	21.7
Server Room	25.5
Class-5	25.7
Class-2	25.9
Class-4	29.5
Class-1	29.8
Computer Lab	33.0
Library	35.7

ALL ROOMS — Ranked by Acoustic Distance

Room name	Acoustic Distance (Hops)
Lobby	5.5
Storage	6.9
Technical Room	8.2
Hall	11.7
Administration	12.8

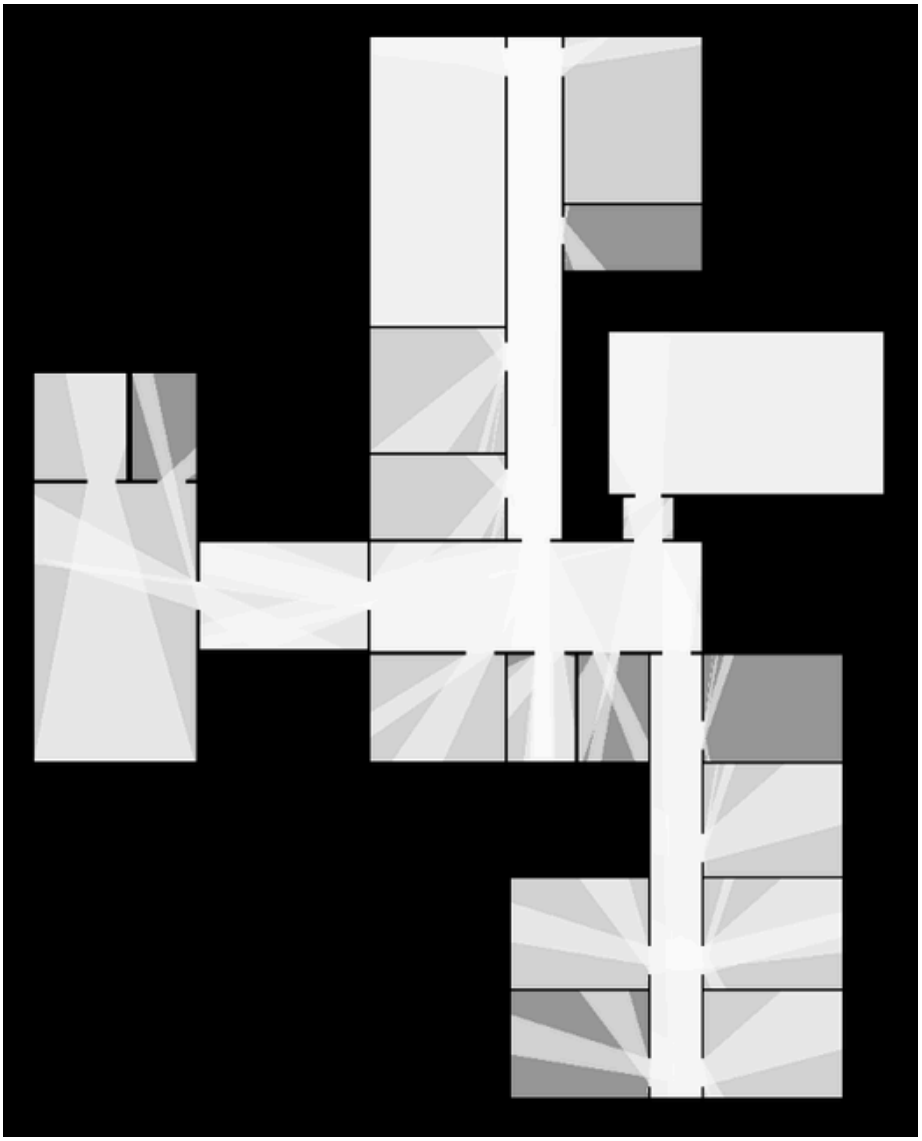
TOP 5 — Most Acoustically Exposed
(non-source rooms closest to noise)

Room name	Acoustic Distance (Hops)
Class-2	25.9
Class-4	29.5
Class-1	29.8
Computer Lab	33.0
Library	35.7

TOP 5 — Most Acoustically Protected
(furthest from all noise sources)

Interpretation

The classrooms and Library score the highest acoustic distances (24–38 hops), confirming they are the most protected spaces. The Hall and Corridor 1 and 2 act as acoustic highways, as they jave high betweenness centrality shows that virtually all sound paths between the noisy public zone and the quiet classroom wing pass through them. The Cafeteria, despite being a major noise source, is naturally buffered by the Lobby, its narrow connection to the main building limits how much sound leaks into the central Hall.



Isovist Analysis
Visibility as acoustic proxy